2 SOURCES AT ANGULAR POSITION ξ_a, ξ_b
$(\xi_a - \xi_b = \Theta)$

→ MICHELSON: 2 SLITS AT SEPARATION $d = d_1 - d_2$
DETECTOR AT POSITION δ

→ HBT : 2 DETECTORS AT SEPARATION d

<u>DISTANCES</u> FROM OBJECT a, b TO SLITS/DETECTORS $1, 2$

- BETWEEN $1,2 \rightarrow \Delta l = \xi_i d$
$a, b \rightarrow \Delta l = \Theta d_i$

- FROM SLITS TO DETECTR $\rightarrow \Delta l = d\delta$

⇒ MICHELSON
SOURCE i, SLIT j, DETECTOR AT δ
$l = (\xi_i + \delta) d_j$

⇒ HBT
SOURCE i, DETECTOR j
$l = \xi_i d_j$

<u>HELPERS:</u>

$$F(x) = \int \tilde{F}(k)\, e^{ikx}\, dk$$

$$|F(x)|^2 = \iint \tilde{F}(k)\, \tilde{F}^*(k')\, e^{i(k-k')x}\, dk\, dk'$$

<u>FIELDS:</u>

$$E_a^i = \alpha\, e^{ikl_{ai}\cos\phi}\, e^{i\omega t} \rightarrow \langle e^{i\cos\phi} \rangle = 0$$
$$\langle e^{i2\cos\phi} \rangle = 0$$

$$E_b^i = \beta\, e^{ikl_{bi}}$$

$$I_a = |\alpha|^2 \qquad V_0 = \frac{I_a - I_b}{I_a + I_b}$$
$$I_b = |\beta|^2$$

②

MICHELSON

$$I(\alpha) = (E_a + E_b)(E_a^* + E_b^*)$$

$$= (E_a^{(1)} + E_b^{(1)} + E_a^{(2)} + E_b^{(2)})(E^* + E^* + E^* + E^*)$$

$$\langle I(\delta) \rangle = \langle |E_a^{(1)}|^2 \rangle + \langle |E_a^{(2)}|^2 \rangle + \langle |E_b^{(1)}|^2 \rangle + \langle |E_b^{(2)}|^2 \rangle$$

$$+ \langle E_a^{(1)} E_a^{(2)*} + E_a^{(2)} E_a^{(1)*} \rangle$$
$$+ \langle E_b^{(1)} E_b^{(2)*} + E_b^{(2)} E_b^{(1)*} \rangle$$

OTHER 8 TERMS CANCEL DUE TO $\Delta\phi$

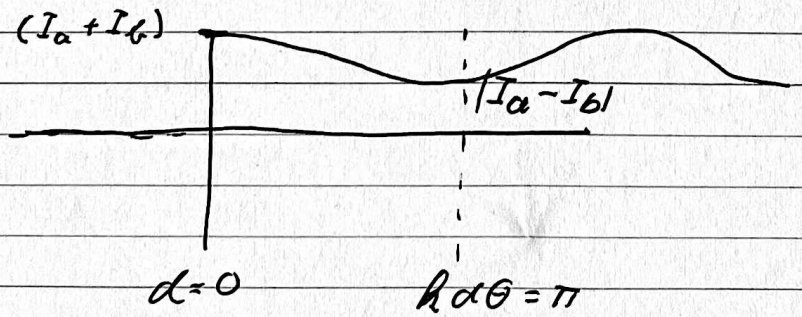
$$= 2I_a + 2I_b + 2I_a \, Re\{ e^{ikd(\delta_a + \frac{1}{2}\delta\alpha)} \}_{+\delta\alpha)}$$
$$+ 2I_b \, Re\{ e^{ikd\delta_b\alpha} \}$$

$$= 2(I_a + I_b) + 2I_a \cos(kd(\delta_a + \delta)) + 2I_b \cos(kd(\delta_b + \delta))$$

$$= 2(I_a + I_b) + 2\sqrt{I_a^2 + I_b^2 + 2I_a I_b \cos(kd\Theta)} \cos(kd\delta + \phi)$$

$$(\Theta = \delta_a - \delta_b)$$

⇒ FRINGE AMPLITUDE AS FUNCTION OF δ



⇒ $\langle I(\delta) \rangle$ OSCILLATES FOR DET. POSITION δ

THE AMPLITUDE OSCILLATES FOR SLIT SEPARATION d

MICHELSON EXTENDED SOURCE

$$\langle I(\delta) \rangle = \int I(\xi) d\xi + \int I(\xi) \cos(k d(\xi + \delta))$$

$$= I_{TOT} + \int I(\xi) \cos(k d\xi) \cos(k d\delta)$$

$$\int I(\xi) \sin(k d\xi) \sin(k d\delta)$$

CALC: $C(d) = \int I(\xi) \cos(k d\xi)$

$$S(d) = \int I(\xi) \sin(k d\xi)$$

$$\langle I(\delta) \rangle = I_{TOT} + C \cos(k d\delta) + S \sin(k d\delta)$$

$$= I_{TOT} + \sqrt{C^2 + S^2} \cos(k d\delta + \phi)$$

AMPLITUDE: $A(d) = \sqrt{C^2(d) + S^2(d)}$

FOR EVEN/SYMMETRIC OBJECT $S = 0$

$$A(d) = C(d)$$

IN GENERAL

① FROM $I(d)$ FRINGE AMPLITUDE AND PHASE DERIVE $C(d)$, $S(d)$

$$\text{BRIGHTNESS}(\xi) = \int (C(d) + i S(d)) e^{-k d\xi} dd$$

$$\left(C(d) + i S(d) = \int B(\theta) e^{i k d\theta} d\theta \right)$$

⇒ EVERY d GIVES YOU ONE FOURIER COMPONENT FOR WAVENUMBER kd

HBT: MEASURE $I_1(\alpha_1)$ $I_2(\alpha_1)$

WE COMPARE $\langle I_1 \rangle^2$ $\langle I_1^2 \rangle$ $\langle I_2 \rangle^2$ $\langle I_2^2 \rangle$ $\langle I_1 I_2 \rangle$

$$\Rightarrow I_i = (E_a^i + E_b^i)(E_a^{i*} + E_b^{i*})$$

$$= |E_a^i|^2 + |E_b^i|^2 + 2\,\mathrm{Re}\{E_a^i E_b^{i*}\}$$

$$\langle I_i \rangle = \langle I_a^i \rangle + \langle I_b^i \rangle$$

$$= |\alpha|^2 + |\beta|^2$$

$$\Rightarrow I_i^2 = (E_a^i + E_b^i)(E_a^{i*} + E_b^{i*})(E_a + E_b)(E_a^* + E_b^*)$$

= 16 TERMS: ONLY TERMS FULLY CONSISTING OF $E_x^i E_x^{i*}$ SURVIVE EASEMBLE AVERAGE $\Rightarrow$ 6 TERMS LEFT

$$\langle I_i^2 \rangle = \langle I_a^{i2} \rangle + \langle I_b^{i2} \rangle + 4 \langle I_a^i \rangle \langle I_b^i \rangle$$

$$= \langle I_a^{i2} \rangle + \langle I_b^{i2} \rangle + 4 \langle I_a^i \rangle \langle I_b^i \rangle$$

CHAOTIC LIGHT GAUSSIAN STATISTICS

$$\Rightarrow \langle I_a^{i2} \rangle = 2 \langle I_a^i \rangle^2$$

$$\Rightarrow \langle I_i^2 \rangle = 2 (\langle I_a^i \rangle + \langle I_b^i \rangle)^2$$

$$= 2 \langle I_i \rangle^2 \Rightarrow \text{SUM IS CHAOTIC} \checkmark$$

$\Rightarrow$ NEXT: CROSS INTENSITY $I_1 I_2$

$$I_1 I_2 = (E_a' + E_b')\cdot(E_a'^* + E_b'^*) + (E_a'^2 + E_b'^2)\cdot(E_a'^{2*} + E_b'^{2*})$$

$= 16$ TERMS
- 8 BECOME 0 IN $\langle .. \rangle$ BECAUSE 3:1 RATIO
- 2 BECOME 0 IN $\langle .. \rangle$ BECAUSE $a^* a^* b b$, $a a b^* b^*$

$\Rightarrow$ 6 TERMS LEFT

$$\langle I_1 I_2 \rangle = \langle I_a'^2 \rangle + I_b$$

$$\langle I_a' I_a^2 \rangle + \langle I_b' I_b^2 \rangle$$

$$+ \langle I_a' I_b^2 \rangle + \langle I_b' I_a^2 \rangle$$

$$+ \langle E_a' E_a^{2*} E_b'^* E_b^{2*} + E_a'^* E_a^{2*} E_b' E_b^2 \rangle$$

ONLY LAST 2 TERMS HAVE A PHASE DEPENDECY (AND POSITION DEPENDENCY)

$$\langle I_1 I_2 \rangle = \ldots\ldots$$

$$+ \langle e^{i(a_1 + a_2 - b_1 - b_2)k} + e^{-i(\ldots)} \rangle |\alpha|^2 |\beta|^2$$

$$= \ldots\ldots$$

$$+ \langle I_a \rangle \langle I_b \rangle 2 \langle 1 \, Re\{ e^{i \ldots} \} \rangle$$

$$= \ldots\ldots$$

$$+ \langle I_a \rangle \langle I_b \rangle \cdot 2 \cos(k(a_1 + a_2 - b_1 - b_2))$$

$$= \ldots$$

$$+ \langle I_a \rangle \langle I_b \rangle 2 \cos(k d \Theta)$$

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$$\langle I_1 I_2 \rangle = \langle (I_a' + I_b')(I_a^2 + I_b^2) \rangle$$

("NAIVE TERM FROM ADDING INTENSITIES")

$$+ 2 \langle I_a' \rangle \langle I_b' \rangle \cos(k d\theta)$$

CHAOTIC LIGHT, EQUAL DETECTION

$$\Rightarrow \langle I_{a1} \rangle = \langle I_{a2} \rangle = I_a$$
$$\langle I_{b1} \rangle = \langle I_{b2} \rangle = I_b$$
$$\langle I_{a1} I_{a2} \rangle = \langle I_{a1} I_{a1} \rangle = \langle I_{a2} I_{a2} \rangle = 2 I_a^2$$
$$\langle I_{b1} I_{b2} \rangle = \langle I_{b0}^2 \rangle = 2 I_b^2$$

$$\frac{\langle I_1 I_2 \rangle}{\langle I_1 \rangle \langle I_2 \rangle} = \frac{\langle I_1 I_2 \rangle}{(I_a + I_b)^2} + \cdots$$

$$= 1 + \frac{I_a^2 + I_b^2}{(I_a + I_b)^2} + \cdots$$

$$g^{(2)}(d) = 1 + \frac{I_a^2 + I_b^2}{(I_a + I_b)^2} + 2\cos(k d\theta) \cdot \frac{I_a I_b}{(I_a + I_b)^2}$$

EQUALLY BRIGHT SOURCES

$$I_a = I_b$$

$$g^{(2)}(d) = 1 + \frac{1}{2} + 2 \cdot \frac{1}{4} \cos(k d\theta)$$

$$= \frac{3}{2} + \frac{1}{2} \cos(k d\theta)$$

$$\boxed{g^2(d) = 1 + \cos^2(k d\theta/2)}$$

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EXTENDED SOURCE
EVERY PAIR ξ_1, ξ_2 CONTRIBUTES

$$\langle I_1 \rangle = \int B(\xi)\, d\xi = \langle I_2 \rangle = I_0$$

$$g^2(d) = \ldots \pm \frac{1}{I_1 I_2} \iint B(\xi) B(\xi') \cos(k d(\xi - \xi'))$$

ASIDE: $\iint B(\xi) B(\xi') \cos(k d(\xi - \xi'))$

$$= \left(\int B(\xi) \cos(k d \xi) \right)^2 + \left(\int B(\xi) \sin(k d \xi) \right)^2$$

$$= C^2(d) + S^2(d)$$

$$g^2(d) = 1 + \frac{C^2(d) + S^2(d)}{I_0^2}$$

$C^2(d) + S^2(d)$:

$$\left| \int B(\xi) e^{ik\xi d}\, d\xi \right|^2$$

$$= \iint B(\xi) B(\xi') e^{ik d(\xi - \xi')}\, d\xi\, d\xi$$

$$= \left| \int B(\xi) \cos(k\xi d) + i \int B(\xi) \sin(k\xi d) \right|^2$$

$$= C^2(d) + S^2(d)$$

$\Rightarrow$ NO INFORMATION ON POSITION
NO INFORMATION ON ORIENTATION
$\Rightarrow$ "BRIGHTNESS AUTOCORRELATION"

BRIGHTNESS AUTO CORRELATION

$$\int B(\rho) B(\rho - \Delta\rho)\, d\rho = A(\Delta\rho)$$

$$g^2(\alpha) = 1 + \frac{|F(\alpha)|^2}{I_0^2}$$

$$|F(\alpha)|^2 = \left| \int B(\rho)\, e^{i k \rho \alpha}\, d\rho \right|^2$$

$$= \iint B(\rho) B(\rho')\, e^{i k \alpha (\rho - \rho')}\, d\rho\, d\rho'$$

$$= \int A(\Delta\rho)\, e^{i k \alpha \Delta\rho}\, d\Delta\rho$$

$$g^2(\alpha) = 1 + \frac{\mathcal{F}(A(\Delta\rho))}{I_0^2}$$

FIRST MEASURE